SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE CODE: ITA 0606 **COURSE NAME: MACHINE LEARNING**

COURSE OUTCOME COVERED

CO1: *Learning Problems, Perspectives and Issues, Concept Learning, Version Spaces & Candidate Elimination, Inductive Bias, Decision Tree Learning, Representation & Heuristic Search (BL1, BL2)*

Assessment Tool Weightage: 40%

QUESTIONS: 5 **Total Marks: 25**

Annexure A

APPLICATION BASED QUESTIONS

Code of Conduct:

I Deepan Kumar P (192312498) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:
Deepan Kumar P

1. Examine the performance of a machine-learning model using its training and validation loss values over multiple epochs: (5 Marks)

(a) Epoch 1: Training Loss = 2.8, Validation Loss = 2.4
(b) Epoch 2: Training Loss = 2.5, Validation Loss = 2.2
(c) Epoch 3: Training Loss = 2.2, Validation Loss = 2.0
(d) Epoch 4: Training Loss = 2.0, Validation Loss = 2.1
(e) Epoch 5: Training Loss = 1.8, Validation Loss = 2.3
(f) Epoch 6: Training Loss = 1.6, Validation Loss = 2.6
(g) Epoch 7: Training Loss = 1.5, Validation Loss = 2.9
(h) Epoch 8: Training Loss = 1.3, Validation Loss = 3.2

i) Identify the epoch at which the model begins to overfit. ii) Suggest two effective techniques to reduce overfitting in this model. iii) Discuss how overfitting affects the model's generalization ability on unseen data.

Answer:

In Machine Learning, **loss** is a measure of how far the predicted output is from the actual output. During training, the objective of the model is to minimize the loss function.

- **Training Loss** measures the error on the training dataset.
- **Validation Loss** measures the error on unseen (validation) data.

A well-trained model should show a decrease in both training and validation loss. However, when the **training loss continues to decrease while the validation loss begins to increase**, the model starts memorizing the training data instead of learning general patterns. This phenomenon is known as **overfitting**.Overfitting reduces the model's ability to make accurate predictions on new, unseen data.

Observation

From the given data:

- **Training loss decreases continuously** from **2.8** to **1.3**, indicating that the model is learning the training data effectively.
- **Validation loss decreases only until Epoch 3**, reaching its minimum value of **2.0**.
- From **Epoch 4 onwards**, validation loss starts increasing (**2.1 → 3.2**) even though training loss keeps decreasing.

This indicates that after Epoch 3, the model begins to memorize the training data instead of learning generalized features.

(i) Identify the epoch at which the model begins to overfit.

Answer

The model **begins to overfit at Epoch 4**.

Justification

- At **Epoch 3**, the validation loss is at its lowest value (**2.0**).
- Beginning with **Epoch 4**, validation loss increases while training loss continues to decrease.

This divergence is the classic indication of overfitting.

(ii) Suggest two effective techniques to reduce overfitting in this model.

1. Early Stopping

Early Stopping is a regularization technique that stops the training process when the validation loss starts increasing.

Advantages

- Prevents the model from memorizing training data.
- Saves computational time.
- Produces better generalization.

In this case, the training should ideally stop after **Epoch 3**, where validation loss is minimum.

(iii) Discuss how overfitting affects the model's generalization ability on unseen data.

Overfitting causes the model to learn the training data too precisely, including noise and unnecessary details.

As a result:

- Training loss becomes very low.
- Validation loss increases.
- The model performs excellently on training data.
- Prediction accuracy decreases on unseen test data.
- The model becomes less reliable for real-world applications.

For example, in medical diagnosis or industrial quality inspection, an overfitted model may fail to correctly classify new patient records or defective products, reducing its practical usefulness.

2. Examine the performance of a machine-learning model using training and validation accuracy over multiple epochs: (5 Marks)

(a) Epoch 1: Training Accuracy = 60%, Validation Accuracy = 58%
(b) Epoch 2: Training Accuracy = 65%, Validation Accuracy = 63% (c) Epoch 3: Training Accuracy = 70%, Validation Accuracy = 68%
(d) Epoch 4: Training Accuracy = 75%, Validation Accuracy = 72%
(e) Epoch 5: Training Accuracy = 80%, Validation Accuracy = 73%
(f) Epoch 6: Training Accuracy = 85%, Validation Accuracy = 72%
(g) Epoch 7: Training Accuracy = 88%, Validation Accuracy = 70% (h)Epoch 8: Training Accuracy = 92%,Validation Accuracy = 68% i) Identify the epoch at which the model starts overfitting.

ii) Explain why validation accuracy decreases despite training accuracy increasing. iii) Suggest two methods to improve validation performance.

<u>Answer:</u>

In Machine Learning, **accuracy** is the percentage of correctly classified samples.

- **Training Accuracy** indicates how well the model performs on the training dataset.
- **Validation Accuracy** measures the model's performance on unseen data.

An ideal model should have both training and validation accuracy increasing together. If the training accuracy continues to increase while validation accuracy starts decreasing, it indicates that the model is learning the training data too specifically instead of learning general patterns. This phenomenon is called **overfitting**.

Observation

From the given data:

- Training accuracy continuously increases from **60% to 92%**.
- Validation accuracy increases only up to **73% at Epoch 5**.

- After **Epoch 5**, validation accuracy starts decreasing (**72%, 70%, 68%**) even though training accuracy keeps increasing.

This difference indicates that the model is no longer generalizing well and has started overfitting.

(i)

Epoch	Training Accuracy	Validation Accuracy	Observation
1	60%	58%	Learning begins
2	65%	63%	Both accuracies improve
3	70%	68%	Good learning
4	75%	72%	Good generalization
5	**80%**	**73%**	**Highest validation accuracy**
6	**85%**	**72%**	**Validation accuracy decreases → Overfitting begins**
7	88%	70%	Overfitting continues
8	92%	68%	Severe overfitting

(ii) Explain why validation accuracy decreases despite training accuracy increasing.

Training accuracy increases because the model continues learning the training dataset and gradually memorizes its patterns.

However, after a certain point, the model also starts memorizing **noise, outliers, and unnecessary details** instead of learning generalized features.

As a result:

- Training accuracy continues to improve.
- Validation accuracy decreases because the model performs poorly on unseen data.

This phenomenon is called **overfitting**, where the model fits the training data extremely well but loses its ability to generalize to new data.

(iii) Suggest two methods to improve validation performance.

1. Early Stopping

Early Stopping halts the training process when the validation accuracy stops improving or validation loss starts increasing.

Benefits

- Prevents overfitting.
- Saves training time.
- Improves model generalization.

In this example, the training should ideally stop at **Epoch 5**, where validation accuracy reaches its maximum value.

2. Data Augmentation

Data Augmentation artificially increases the size and diversity of the training dataset by applying transformations such as:

- Rotation
- Flipping
- Cropping
- Scaling

- Noise addition

Benefits

- Increases dataset diversity.
- Reduces overfitting.
- Improves validation accuracy.
- Enhances the model's ability to generalize to unseen data.

3. In an industrial process, an AI robot is assigned to segregate the front and back tires automatically. Let the number of tires are 250 (125 front and 125 back tires). (5 Marks)

a) Front identified as Front: 119
b) Front identifies as Back: 06
c) Back identified as Back: 120
d) Back identified as Front: 05

Compute: Accuracy, Precision, Sensitivity, Specificity and F1-Score.

Code:

```
# ----------------------------------------
# AI Tire Classification Performance Metrics
# ----------------------------------------

# Confusion Matrix Values
TP = 119   # Front identified as Front
FN = 6     # Front identified as Back
TN = 120   # Back identified as Back
FP = 5     # Back identified as Front

# Total Samples
total = TP + TN + FP + FN

# Accuracy
accuracy = (TP + TN) / total

# Precision
precision = TP / (TP + FP)

# Sensitivity (Recall)
recall = TP / (TP + FN)

# Specificity
specificity = TN / (TN + FP)

# F1-Score
f1_score = (2 * precision * recall) / (precision + recall)

# Display Results
print("====== AI Tire Classification Performance ======\n")
```

```
print(f"True Positive (TP)  : {TP}")
print(f"False Positive (FP) : {FP}")
print(f"True Negative (TN)  : {TN}")
print(f"False Negative (FN) : {FN}\n")

print(f"Accuracy           : {accuracy*100:.2f}%")
print(f"Precision          : {precision*100:.2f}%")
print(f"Sensitivity (Recall): {recall*100:.2f}%")
print(f"Specificity        : {specificity*100:.2f}%")
print(f"F1-Score           : {f1_score*100:.2f}%")
```

```
===== AI Tire Classification Performance ======

True Positive (TP)  : 119
False Positive (FP) : 5
True Negative (TN)  : 120
False Negative (FN) : 6

Accuracy            : 95.60%
Precision           : 95.97%
Sensitivity (Recall): 95.20%
Specificity         : 96.00%
F1-Score            : 95.58%
```

4.In a medical diagnosis system, an AI model detects whether a patient has a disease or not. Out of 300 cases (150 diseased and 150 healthy): (5 Marks)

a) Diseased identified as Diseased: 138
b) Diseased identified as Healthy: 12
c) Healthy identified as Healthy: 135
d) Healthy identified as Diseased: 15

Compute: Accuracy, Precision, Sensitivity (Recall), Specificity, and F1-Score.

Answer:

```
# ----------------------------------------
# Medical Diagnosis System Performance Metrics
# ----------------------------------------

# Confusion Matrix Values
TP = 138   # Diseased identified as Diseased
FN = 12    # Diseased identified as Healthy
TN = 135   # Healthy identified as Healthy
FP = 15    # Healthy identified as Diseased

# Total Cases
total = TP + TN + FP + FN
```

```
# Accuracy
accuracy = (TP + TN) / total

# Precision
precision = TP / (TP + FP)

# Sensitivity (Recall)
recall = TP / (TP + FN)

# Specificity
specificity = TN / (TN + FP)

# F1-Score
f1_score = (2 * precision * recall) / (precision + recall)

# Display Results
print("====== Medical Diagnosis System Performance ======\n")

print(f"True Positive (TP)  : {TP}")
print(f"False Positive (FP) : {FP}")
print(f"True Negative (TN)  : {TN}")
print(f"False Negative (FN) : {FN}\n")

print(f"Accuracy            : {accuracy*100:.2f}%")
print(f"Precision           : {precision*100:.2f}%")
print(f"Sensitivity (Recall): {recall*100:.2f}%")
print(f"Specificity         : {specificity*100:.2f}%")
print(f"F1-Score            : {f1_score*100:.2f}%")
```

```
====== Medical Diagnosis System Performance ======

True Positive (TP)  : 138
False Positive (FP) : 15
True Negative (TN)  : 135
False Negative (FN) : 12

Accuracy            : 91.00%
Precision           : 90.20%
Sensitivity (Recall): 92.00%
Specificity         : 90.00%
F1-Score            : 91.09%
```

5.A medical image classification system using a deep learning framework combined with K-Nearest Neighbor (KNN) yields TP = 180 and TN = 160. Each class contains 1200 samples, and 25% of the dataset is used as test data.

Calculate: Accuracy Precision Sensitivity Specificity. (5 Marks)

Answer:

```
# ---------------------------------------
# Medical Image Classification Performance
# ---------------------------------------

# Given Values
TP = 180
TN = 160

# Total samples
samples_per_class = 1200
total_samples = samples_per_class * 2

# 25% used for testing
test_samples = int(total_samples * 0.25)

# Test samples in each class
positive = test_samples // 2
negative = test_samples // 2

# Calculate FP and FN
FN = positive - TP
FP = negative - TN

# Performance Metrics
accuracy = (TP + TN) / test_samples
precision = TP / (TP + FP)
recall = TP / (TP + FN)
specificity = TN / (TN + FP)

# Display
print("====== Medical Image Classification ======\n")

print(f"Total Dataset          : {total_samples}")
print(f"Test Samples           : {test_samples}")
print(f"Positive Test Samples  : {positive}")
print(f"Negative Test Samples  : {negative}\n")

print(f"TP : {TP}")
print(f"TN : {TN}")
print(f"FP : {FP}")
```

```
print(f"FN : {FN}\n")

print(f"Accuracy     : {accuracy*100:.2f}%")
print(f"Precision    : {precision*100:.2f}%")
print(f"Sensitivity  : {recall*100:.2f}%")
print(f"Specificity  : {specificity*100:.2f}%")
```

```
••• ====== Medical Image Classification ======

    Total Dataset          : 2400
    Test Samples           : 600
    Positive Test Samples  : 300
    Negative Test Samples  : 300

    TP : 180
    TN : 160
    FP : 140
    FN : 120

    Accuracy     : 56.67%
    Precision    : 56.25%
    Sensitivity  : 60.00%
    Specificity  : 53.33%
```

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	**25**	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	**25**	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	**20**	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	**15**	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	**15**	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand